UNIT 5 · TOPIC 5(A)

Safety Approach, Tractor Accidents and ROPS

Syllabus topic: Safety aspects of agricultural machinery

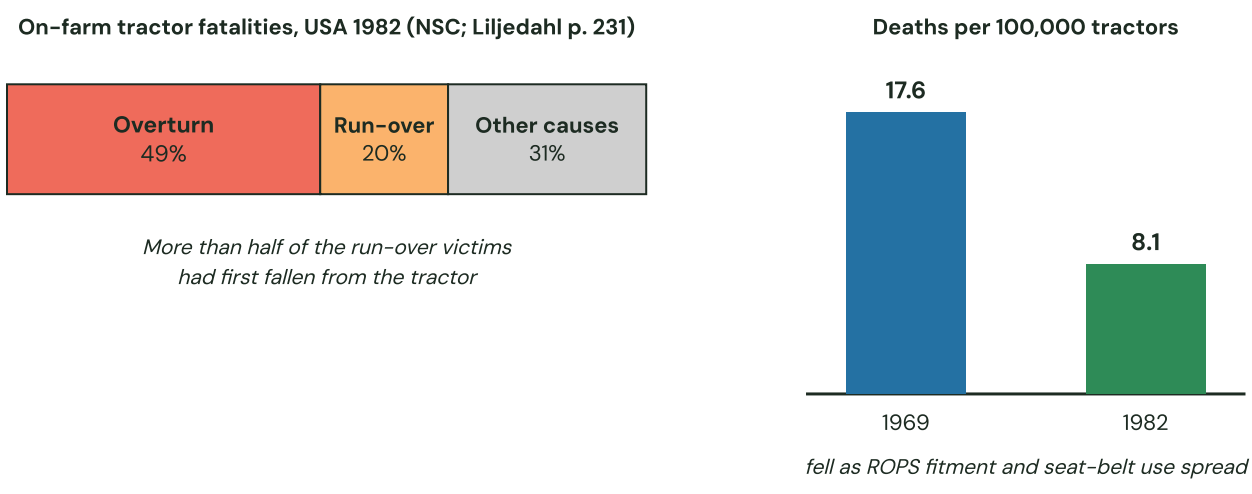
The safety approach (Liljedahl p. 234)

- Safety improves through **EDUCATION**, **REGULATION** (laws) and **DESIGN**.
- All three work, but the most positive method is to design the product or system so the **HAZARD IS REMOVED** — the engineer's core responsibility.
- Design conflicts are real: an inconvenient shield soon gets discarded, and a machine whose function a safety device seriously hampers will not sell (pp. 234–235).
- Good safety design is therefore convenient, functional and unobtrusive.

Hazard-control hierarchy (the modern framing of the same idea):

1. ELIMINATE / SUBSTITUTE.
2. ENGINEERING CONTROLS (guards, interlocks, ROPS).
3. ADMINISTRATIVE (training, signage, intervals).
4. PPE — the LAST line of defence (Topic 8).

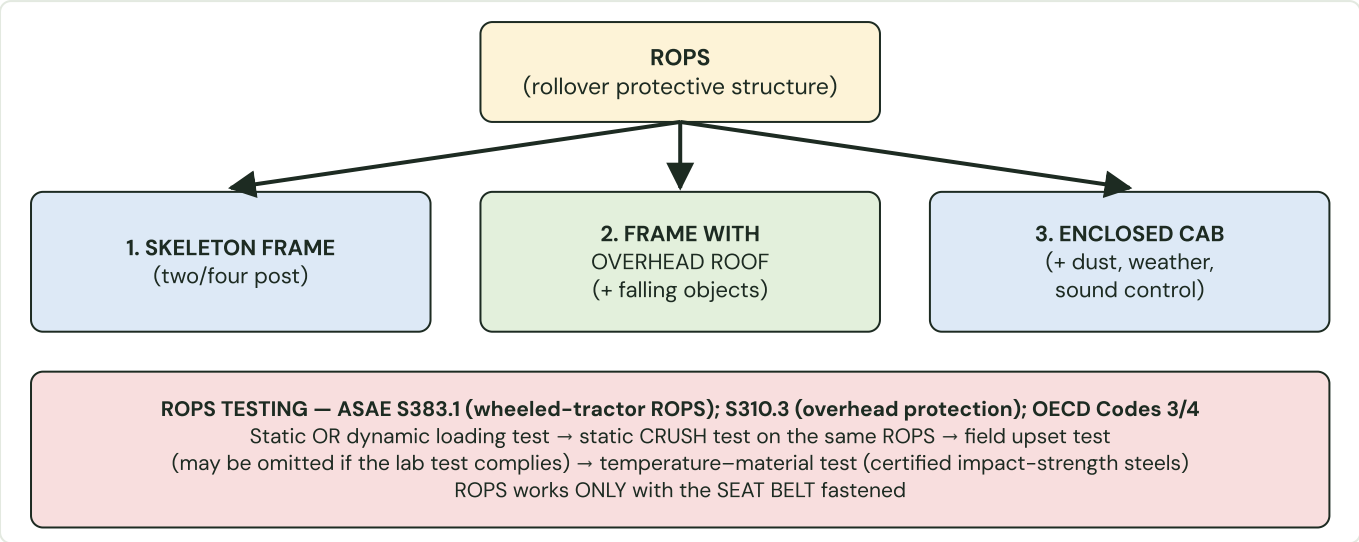
Tractor accidents (US data, p. 231)



US tractor-fatality data (redrawn using only the values stated in the text)

- US National Safety Council, 1982: **OVERTURNS** caused **49%** of on-farm tractor fatalities; **20%** were run-over deaths — in more than half of those the victim had first FALLEN FROM the tractor (the case against riders/passengers).
- From 1969 to 1982 deaths per 100,000 tractors fell from **17.6 to 8.1** as ROPS fitment and seat-belt use spread.

Rollover protective structures (ROPS, p. 232)



ROPS types and testing (redrawn from the source figure)

ROPS type	Protects against
1. Skeleton frame	Overturn
2. Frame with overhead roof	Overturn + falling objects
3. Enclosed cab	Overturn + falling objects; also gives environmental and sound control

Standards: ASAE S383.1 (ROPS for wheeled agricultural tractors) and S310.3 (overhead protection); OECD codes internationally (Unit 4, Topic 11).

Testing sequence:

1. Static OR dynamic loading test in the laboratory.
2. Static CRUSH test on the same ROPS.
3. Field upset test (may be omitted if the lab test shows compliance).
4. Temperature–material test, using certified impact-strength steels.

- Design analysis today uses finite-element simulation to predict the yielding mode and check welds/joints (p. 232).

> **CRITICAL:** ROPS protects only when the SEAT BELT is worn — the belt keeps the operator inside the protected zone.

Overturn mechanics (recap — Unit 2, Topic 4; Unit 4, Topic 9a)

- Lateral upset: the CG vertical passes outside the downhill wheels — $\tan\theta = (B/2)/h$.
- Rearward upset: high hitching + torque reaction.
- **Prevention:** low drawbar hitching only, wide track, ballast, slow on slopes and ditch banks, no sharp turns at speed, ROPS + belt.

Practice MCQs

Q71. Per Liljedahl, the MOST positive method of improving safety is:

(a) Education
(b) Regulation
(c) Designing the hazard out of the product
(d) Insurance

Q72. A safety shield that is inconvenient will most likely be:

(a) Praised
(b) Discarded by the operator
(c) Mandatory
(d) Cheaper

Q73. In US 1982 data, overturns accounted for what share of on-farm tractor fatalities?

(a) 20%
(b) 33%
(c) 49%
(d) 75%

Q74. Run-over deaths accounted for about:

(a) 5%
(b) 20%
(c) 40%
(d) 60% of tractor fatalities

Q75. In more than half of run-over cases the victim had first:

(a) Fallen from the tractor
(b) Been asleep
(c) Been on foot far away
(d) Been in a cab

Q76. Deaths per 100,000 tractors fell from 17.6 (1969) to what value in 1982?

(a) 15.2
(b) 12.0
(c) 8.1
(d) 2.0

Q77. That fall is attributed mainly to:

(a) Slower tractors
(b) Increasing ROPS fitment and seat-belt use
(c) Fewer tractors
(d) Better roads

Q78. Which is NOT one of the three basic ROPS types?

(a) Skeleton frame
(b) Frame with overhead roof
(c) Enclosed cab
(d) Roll cage of fabric

Q79. ROPS for wheeled agricultural tractors is covered by:

(a) ASAE S383.1
(b) ASAE S209
(c) ISO 500
(d) IS 6284

Q80. Overhead protection test procedures are covered by:

(a) S310.3
(b) J833
(c) IS 4931
(d) ISO 730

Q81. In ROPS testing, the static or dynamic loading test is followed by a:

(a) Paint test
(b) Static crush test on the same ROPS
(c) Fuel test
(d) Noise test

Q82. The field upset test may be omitted if:

(a) The tractor is small
(b) The laboratory test indicates compliance
(c) It rains
(d) Never

Q83. ROPS protects the operator ONLY if:

(a) The cab is closed
(b) The seat belt is fastened
(c) Lights are on
(d) Speed is high

Q84. Besides overturn, ROPS with roof protects against:

(a) Falling objects
(b) Noise only
(c) Fuel loss
(d) Dust only

Answers

Q71 — (c) Designing the hazard out of the product — *Liljedahl p. 234 — design that removes the hazard is the most positive method.*

Q72 — (b) Discarded by the operator — *pp. 234–235 — an inconvenient shield is soon discarded.*

Q73 — (c) 49% — *p. 231 (NSC Accident Facts 1983, for 1982).*

Q74 — (b) 20%

Q75 — (a) Fallen from the tractor — *More than half had fallen from the tractor first.*

Q76 — (c) 8.1 — *Rate 17.6 → 8.1 per 100,000 tractors.*

Q77 — (b) Increasing ROPS fitment and seat-belt use

Q78 — (d) Roll cage of fabric — *p. 232 — the three ROPS types.*

Q79 — (a) ASAE S383.1

Q80 — (a) S310.3

Q81 — (b) Static crush test on the same ROPS — *p. 232 — static or dynamic → crush → field upset (omittable) → temperature–material.*

Q82 — (b) The laboratory test indicates compliance

Q83 — (b) The seat belt is fastened

Q84 — (a) Falling objects

Sources: Liljedahl et al. — Tractors and their Power Units, Ch. 9, pp. 231–235; ASAE S383.1 & S310.3; BIS safety standards IS 9020 / IS 9129 / IS 12239; Insecticides Act 1968 & Rules 1971 (label classes); WHO hazard classification; ICAR-CIAE / AICRP farm-accident surveys; NPTEL lectures.